

70
Yb

Ytterbium

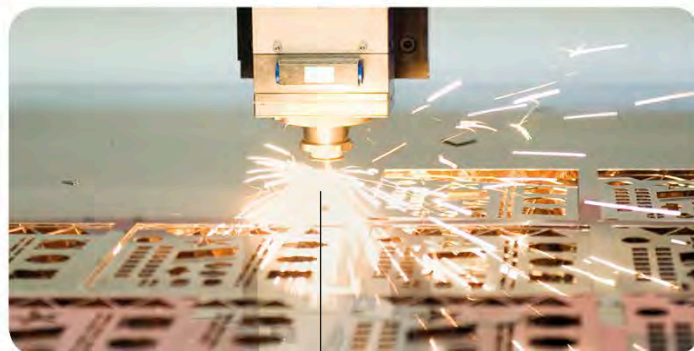
Ytterbium tends to be more reactive than other lanthanide metals. It is stored in sealed containers to stop the metal from reacting with oxygen. The **pure metal** has only a few uses. A small amount of ytterbium is used in making steel, while its compounds are used in some **lasers**.

This bright, shiny metal can be hammered into thin sheets.

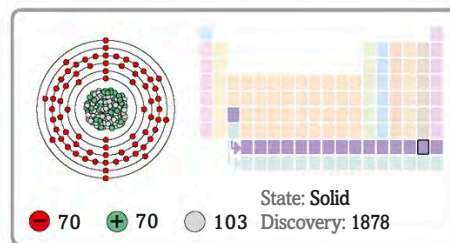
Laboratory sample of pure ytterbium



Laser cutting



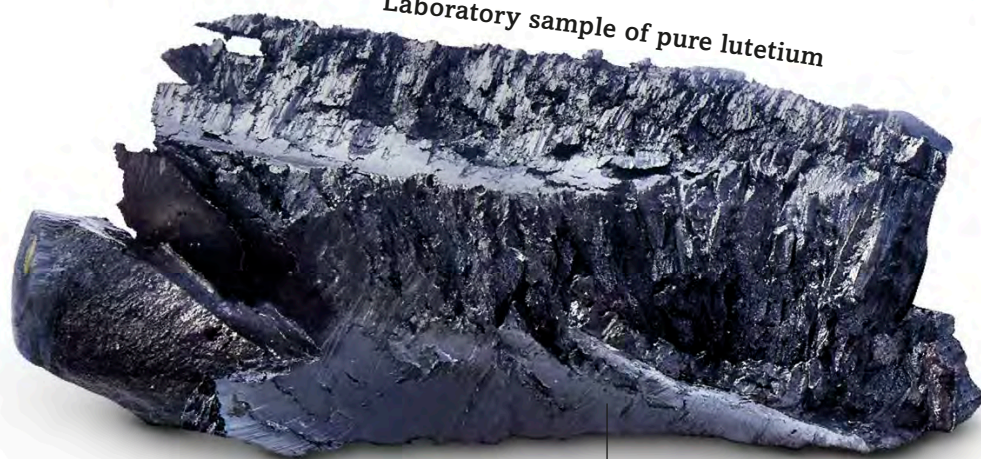
An ytterbium laser can cut through metals and plastics.



71
Lu

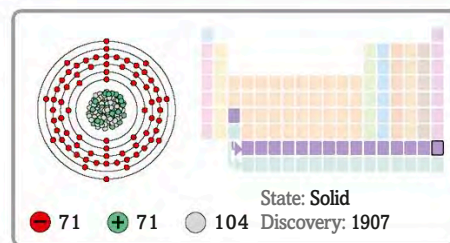
Lutetium

Laboratory sample of pure lutetium



This element is the hardest and densest lanthanide metal.

Lutetium was the last of the rare earth metals to be **discovered**. It is also the final member of the lanthanides. In its **pure form**, lutetium is very reactive and catches fire easily. It is rare and has few uses, mainly as a substance mixed with crude **oil**.



Some oil refineries use lutetium to break down crude oil to make fuels, such as petrol and diesel.

Oil refinery

